# Smart Multi-Sensor Wearable for Real-time Elderly Health Anomaly and Fall Detection

R26-IT-108

Project Proposal Report

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Sri Lanka Institute of Information Technology Sri Lanka

2026 March

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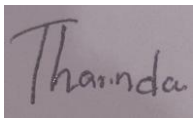
Department of Information Technology

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2026 March

# DECLARATION

I declare that this proposal is my own work and does not incorporate, without acknowledgement, any material previously submitted for a degree or diploma at any university or institute of higher learning. To the best of my knowledge and belief, it does not contain any material previously published or written by another person except where due acknowledgement is made in the text.

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The above candidate is carrying out this research for the undergraduate dissertation under my supervision.

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**Date:** 12.03.2026

# ABSTRACT

The study introduces the design and development of a hip-worn wearable device in real-time to detect falls in elderly individuals and provide post-fall immobility confirmation. Among elderly people falls are a significant health hazard, and may cause severe injuries, delayed treatment, and loss of independence. Even though there are numerous wearable fall detection systems, it is a usual issue that many generate a lot of false alarms because of natural daily activities like sitting, bending, or lying down. Moreover, most systems are only interested in monitoring the fall instance and not checking the user state after the fall. To fill this gap, the proposed system employs a two-stage detection method. The accelerator and gyroscope data of an MPU6050 sensor are used to identify the first, sudden motion and impact patterns. Second, the system verifies the occurrence through body abnormal posture and immobility period following the impact detected. The wearable is constructed with an ESP32 microcontroller and is developed as a belt-worn prototype to guarantee feasible location on the body in the vicinity to the centre of mass. Performance will be measured using public datasets and controlled testing. The anticipated result is the development of a low cost, real time and reliable fall detection feature that enhances precision and minimizes false alarms in elderly monitoring solutions.

# LIST OF ABBREVIATIONS

| Abbreviation | Full Form |
|---|---|
| ESP32 | Espressif 32-bit Microcontroller |
| MPU6050 | Motion Processing Unit 6050 |
| IMU | Inertial Measurement Unit |
| IoT | Internet of Things |
| ADL | Activities of Daily Living |
| RVM | Resultant Vector Magnitude |
| SMA | Signal Magnitude Area |
| CNN | Convolutional Neural Network |
| ML | Machine Learning |
| USB | Universal Serial Bus |

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# 1. INTRODUCTION

## 1.1 Background and Context

### *1.1.1 Overview of Elderly Fall Risks*

Falls are a health issue among elderly individuals due to their frequent occurrences causing injury, loss of autonomy, lack of confidence regarding mobility, and escalated long-term care demand. As individuals grow older, balance, gait stability, muscle strength, reaction time, and postural control decline, which predispose them to accidental falls in their day-to-day activities [1], [4]. Besides causing physical injury, falls may have psychological effects including fear of falling, which subsequently may lead to lower levels of activity and increased functional decline with time [1]. These consequences make fall-related incidents one of the major concerns in aged care, especially in people who live alone or need constant supervision.

When a fall is accompanied by a long period of immobility, the risk is even more serious. The elderly in these cases might not be able to stand, call, or regain a safe position without any external assistance. It may result in the loss of prompt help, complications, and worse recovery results [7]. This is why the current research on elderly monitoring has gone beyond the primary focus on a single fall detection instrument and has started to focus more on the necessity to detect not only the act of the fall but also the physical state that it leaves [3], [4].

### *1.1.2 Importance of Fall and Immobility Detection*

Adequate fall and immobility detection is significant since early treatment can minimize the extent of post-fall complications. Early detection of the event enables family members, caregivers, or healthcare personnel to intervene in a timely manner and offer help before the condition deteriorates [2], [3]. This is especially significant in-home care for the elderly, where direct supervision is not always present.

The detection of immobility is also significant since not all falls lead to instant movement or self-recovery. A motionless older adult after the impact may need immediate assistance, although the primary fall has ceased [7], [8]. Consequently, an effective monitoring system not

only needs to detect quick body movement or impact in the body, but it must also confirm that the individual is not in abnormal position or inactive position at the end of the action. Such a hybrid solution can enhance the usability of wearable fall monitoring in practice [9], [10].

### *1.1.3 Challenges in Existing Fall Detection Systems*

Despite the numerous fall detection systems proposed, there are still many practical and technical challenges that are present. A significant problem is the inability to differentiate actual falls and daily acts like sitting down sitting down fast, lying on a bed, bending, or changing direction abruptly [2], [11]. The result of this overlap of motion patterns is a frequent false alarm that decreases user trust and system reliability.

The other difficulty is that most of the studies are tested in controlled experimental settings and not in the real world. Algorithms can be trained on public datasets, but real falls are hard to realize in nature and are uncommon, which creates a permanent issue of model generalization [7], [13]. Moreover, a few systems require computationally intensive models or multi-device infrastructures, which might not work effectively with low-cost wearable implementations [3], [12]. Such constraints demonstrate that simple, reliable, and wearable fall detection systems are still required to function in real scenarios in terms of everyday monitoring of the elderly.

### *1.1.4 Need for a Hip-Worn Wearable Belt Solution*

A wearable hip-belt is a good design for fall and immobility detection since the hip area is near the centre of mass of the body and can better detect trunk movement, posture, and impact-related motion in comparison to other body regions [8], [9]. The signal obtained through this area can have an informative value about acceleration, angular velocity, body position, and post-fall inactivity, which will prove useful in identifying fall events.

A belt-based device can also provide a more consistent placement and more stable motion measurements, as compared to loosely positioned or poorly aligned sensors, when used in daily activities. A hip-mounted solution is another option to consider when integrating fall detection with immobility verification since it can monitor sudden body displacement, as well as immobility following the occurrence of such a movement [10]. This kind of solution can offer

a viable trade-off of wearability, detection quality, and real-time functionality to the elderly users [4], [12].

### *1.1.5 Relevance to Elderly Healthcare Monitoring*

Most healthcare is now moving towards preventive, continuous and in-home care. Wearable systems are also significant in this context, as they can be used to identify risk-related events outside hospital settings [1], [4]. Fall detection stands out particularly since it deals with one of the most frequent and dangerous risks of older adults in everyday life.

A relative motion and immobility detection module could be used to further extend the area of elderly monitoring to inform the individual of his or her physical safety in a timely manner. Added to a broader assistive monitoring system, it may assist caregivers with immediate alerts, enhance response preparedness, and supplement other physiological or behavioral surveillance elements [2], [3]. Thus, the current study has a direct bearing on the creation of affordable, wearable, and real-time uses of practical elderly healthcare technologies.

## 1.2 Literature Review

### *1.2.1 Fall Detection in the Elderly Care*

The available literature demonstrates that the fall detection has emerged as a significant field in the elderly assistive technology because of the prevalence rate and the severity of the fall occurrences in the elderly population [1], [2]. The systematic reviews have demonstrated that wearable solutions have been well researched as they are always capable of tracking mobility, yet are portable and relatively cheap [1], [4]. These systems can be modelled to enhance safety in autonomous living conditions and to cut down on the time of emergency response [3].

Meanwhile, the literature suggests that actual implementation is still difficult. Most of the studies have shown encouraging outcomes in the lab, and their performance can be worse when the systems are deployed in the natural day-to-day lives or by a population other than the initial test subjects [7], [13]. This indicates that fall detection in the elderly is a classification problem, as well as practical usability and reliability problem.

### *1.2.2 Wearable Motion IMU-Based Motion Monitoring*

Accelerometers and gyroscopes are wearable inertial measurement units (IMUs) that are the most widely used sensing technology to detect falls as they can record body motions and provide direct data [2], [9]. Acceleration signals are also applicable in detecting sharp impacts and the intensity of movement, but the Gyroscopes data is applicable in detecting rotation and alteration of the body orientation [9], [10]. The combination of these signals will be a kinematic representation of human motion that can be analyzed with regards to falls.

The SisFall, KFall datasets have been used to develop IMU-based methods using fall and activity of daily living labelled data on motion [5], [6]. These datasets have been utilized in testing the traditional algorithms as well as learning-based models and are therefore useful in performance comparisons of detection strategies [5], [6], [10].

### *1.2.2 Hip and Waist Placement for Fall Detection*

The location of sensors can achieve good detection and the hip, or the waist is often tested as a successful site to wear the sensors [8], [9]. Mounted close to the trunk, a sensor can record movements related to body instability, impact, and postural changes more vividly than those mounted at more distant locations that are not close to the core body mass [8]. This especially applies in the fall detection as most falls deal with the rapid transformations in the trunk orientation and acceleration.

Tri-axial accelerometers and gyroscopes have been shown to be effective in supporting threshold-based and orientation-based detection with meaningful results when used as hip- or waist-positioned sensors [9], [10]. These results contribute to the hypothesis that the hip-worn belt is an appropriate physical design solution to this component in a scenario where the objective is to monitor both the incidence of falls and the ensuing inability to move.

### *1.2.3 Threshold-Based and Machine Learning Approaches*

There are two general methodological directions that are typically described in the literature: threshold-based and machine learning approaches. The threshold-based techniques detect fall

through the examination of whether the motion characteristics like acceleration value or angular velocity surpass a set threshold [8], [9]. These algorithms tend to be computationally efficient and can be deployed easily to an embedded system, but false positives may be problematic with normal motion patterns [9].

Machine learning methods are aimed at reducing discrimination through learning trends based on labelled data. Researchers have applied public data sets to categorize falls and day-to-day activities with supervised models and additional improved methods [10], [11]. There are also more recent studies that examine lightweight deep learning solutions to practice wearable systems [12]. Although learning-based approaches can enhance recognition ability, their effectiveness is based on the quality of data, generalization and practicality of implementation on resource-limited devices [11], [12].

### *1.2.5 Edge-based Real-Time Fall detection Systems*

Edge based fall detection is a method to conduct sensing and decision-making at the wearable object or local embedded processor and not at all on external infrastructure. The method is also relevant to real-time tracking since it can minimize latency, enhance privacy, and can be used to execute operations even in situations where the network connectivity is constrained [3], [12]. These advantages are very desirable in practice in monitoring elderly.

Recent research has demonstrated an increase in interest in lightweight models and compact implementations capable of executing on limited wearable hardware [12]. Nevertheless, most of the suggested systems continue to experience trade-offs in terms of accuracy, computational cost, power efficiency, and real-world robustness [3]. Thus, the intent to develop an edge-capable fall detection solution, which is not too expensive and reliable, remains a significant research focus.

## 1.3 Research Gap

### *1.3.1 Limitations of Current Wearable Fall Detection Devices*

The present wearable fall detection devices do illustrate the possibility of continuous monitoring although there are a few limitations. Most of the systems are tested primarily on

simulated falls, with a small number of participants, or under control conditions of activity, which influences the reliability of performance in real life [7], [13]. Moreover, other devices concentrate more on fall detection rather than an adequate focus on what follows the event (either the individual stays in position or he or she gets up) [3], [4].

### *1.3.2 False Alarm Problem in Daily Activity Recognition*

One of the biggest unsolved cases is the problem of false alarm due to mixed up of falls and normal movements. Rapid sitting, lying down, stumbling, falling without falling, or abrupt body turning can give rise to motion signatures that are like falls [2], [9], [11]. False alarms even once a month decrease trust in the system and can lead to decreased use of the system among older adults and caregivers. This demonstrates the necessity of more robust decision logic that would be able to confirm fall events based on more than one impact signal.

### *1.3.3 Gap in Low-Cost Hip-Worn Embedded Detection Solutions*

Even though the use of hip and waist sensors has been explored, affordable embedded designs offering a combination of sensible sensor location, real-time processing, and ease of implementation remain necessary [8], [9], [12]. Other literature focuses on datasets or prototype concepts whereas some studies give priority to algorithmic performance. There are few studies that offer a clear solution that is hip-worn and can be used by elderly people in their daily lives with light weight on-device processing [3], [12].

### *1.3.4 Need for Combined Fall and Immobility Verification*

The literature reviewed indicates that elderly safety monitoring cannot solely be effective by simply detecting a fall. Also required is confirmation of the state of the individual post the incident, specifically, does the abnormal posture and immobility persist beyond a significant period [7], [10]. An integrated fall and immobility cheque device can be used to minimize false alarms and enhance the utility of alerts. As such, this proposal answers the gap by considering a hip-worn wearable belt, which applies kinematic sensing to detect two-stage falls and immobility in a format that could be embedded in real-time to monitor the elderly.

# 2. OBJECTIVE

## 2.1 Main Objective

The main objective of this research is to develop a low-cost, hip-worn wearable motion sensing module for real-time fall and immobility detection in elderly individuals using inertial sensor data and embedded processing. The proposed component is intended to support elderly healthcare monitoring by detecting fall-related kinematic patterns and verifying post-fall inactivity through a practical two-stage detection approach. This objective is aligned with the need for wearable systems that are accurate, responsive, and suitable for continuous daily use in home or assisted-care settings [2], [3], [12].

## 2.2 Specific Objectives

### *2.2.1 Develop a hip-worn motion sensing module*

To design and implement a wearable belt-based sensing module positioned at the hip for monitoring elderly body movement in daily life. The hip location is selected because it is close to the body's center of mass and can provide stable and meaningful motion signals for fall-related analysis [8], [9]. The module will be developed using a compact inertial sensing setup suitable for continuous use.

### *2.2.2 Capture and process accelerometer and gyroscope signals*

To acquire tri-axial accelerometer and gyroscope data related to body movement, posture transitions, and impact events, and to process these signals for fall detection analysis. IMU-based motion monitoring has been widely used in fall detection studies because it provides useful kinematic information for distinguishing falls from daily activities [5], [6], [10]. The processed signals will support feature extraction and decision-making in the proposed system.

#### *2.2.3 Design a two-stage fall detection algorithm*

To develop a two-stage fall detection method that first identifies possible fall events through impact-related motion characteristics and then verifies the event using postural orientation and immobility information. This approach is intended to improve the practical reliability of the system by combining event detection with physical-state confirmation after the incident [9], [10].

#### *2.2.4 Reduce false positives during normal activities*

To reduce false alarms caused by daily activities such as sitting, lying down, bending, or rapid body movement by introducing additional verification logic into the detection process. False positives remain a major challenge in wearable fall detection research, especially under real-life movement conditions [2], [11], [13]. Therefore, this objective focuses on improving system dependability for real-world elderly monitoring.

# 3. METHODOLOGY

## 3.1 Research Design

This proposal gives a design-and-development research design to develop and test a wearable kinematic fall and immobility detection unit to monitor the elderly. The suggested component is deployed as a hip-worn belt device, since hip area offers useful motion data concerning effects on the body, change in posture, and post-fall inactivity [8], [9]. The methodology merges inertial motion sensing, signal processing, lightweight decision logic and embedded implementation to assist real-time fall monitoring in a practical and low-cost implementation. The above-mentioned research design is influenced by previous studies on wearable fall detection, existing benchmark datasets accessible to the public, and in-built real-time monitoring solutions [2], [3], [12].

## 3.2 Hardware and Software Tools

An ESP32 microcontroller and an MPU6050 inertial measurement unit (IMU) will be used to develop the proposed system. The MPU6050 offers tri-axial acceleration and gyroscopes, the tri-axial acceleration and gyroscopes are very popular in fall detection research since it measures both linear and angular movement of the body [9], [10]. The embedded processing platform is the ESP32 whose low cost, small size, and wearable real-time applications make it an appropriate choice. Software tools will consist of the Arduino development environment or embedded programming tools equivalent to firmware implementation, and Python to do offline preprocessing, feature analysis, and performance testing. The design is informed by the wearable and edge-based fall detection studies, and practical information sensing, low complexity and real-time decision-making studies are the guiding factors of the design [2], [12].

## 3.3 Motion Data Acquisition

Motion data acquisition will involve the gathering and processing body movement signals pertinent to falls and post fall immobility. The module will record the tri-axial acceleration and angular velocity of the hip-worn sensor position. It is an appropriate position since it is near the centre of mass of the body and can more accurately represent the action of the trunk and abrupt postural changes [8], [9]. To develop and test algorithms, the research will utilize the available benchmark datasets such as SisFall and KFall, which have included labelled motion data on falls and activities of daily living [5], [6]. These data sets offer a systematic foundation on the study of movement patterns related to impact, change of orientation, and rest.

## 3.4 Feature Extraction and Processing

The motion signals obtained after data acquisition will undergo some processing to extract kinematic features that are pertinent to fall detection. The feature extraction step will consist of measures based on the magnitude of acceleration and body orientation as they are typically used in research of wearable fall detection [9], [10]. Major items should be Resultant Vector

Magnitude (RVM), Signal Magnitude Area (SMA), and indicators of orientation based on accelerometer and gyroscope measurements. Signal processing will also assist in the detection of immobility time after a suspected fall incident. This is significant since thoughtful feature design will potentially enhance the discrimination between actual falls and normal daily tasks that may include sitting down, bending up, or lying down [10], [11].

## 3.5 Two-Stage Fall Detection Method

The two-stage approach to the proposed detection method will be used. At the initial level, the system will detect possible fall incidents using sudden motion alterations and impact-related kinematic features. At this level, threshold-based logic will be appropriate since the effect of impact events tends to create impressive spikes in acceleration or mixed movement indicators [8], [9]. The second stage will involve the system checking whether the identified event is accompanied by abnormal body positioning and immobility duration. This second layer is added to minimize false positives due to normal but sporadic daily events [2], [11], [13]. The proposed method seeks to enhance detection reliability by integrating pre- and post-event physical-state cheques with initial impact recognition to enable the implementation of the proposed method by elderly individuals.

## 3.6 ESP32-Based Implementation

The completed detection logic will be put into practice with the ESP32, such that local sensing and fall decision-making can be done on the wearable device. This integrated design facilitates real-time functionality, minimizes reliance on third party infrastructure, and is more appropriate to low-end elderly surveillance applications [3], [12]. The sensor data will be accommodated in the MPU6050 and sent to ESP32, which will then perform the feature and decision logic and provide the fall-related alert when necessary. The integrated implementation will hence be the working prototype of the proposed hip-worn system.

## 3.7 Performance Evaluation

Benchmark data and prototype-level testing will be used to measure system performance. The primary evaluation criteria will involve the ability to identify falls properly, decrease false alarms when moving daily, and determine post-fall immobility timely. Previous research studies have revealed that false positive rate, sensitivity, and practical robustness are essential in fall detection research [1], [10], [13]. The proposed system will therefore be evaluated based on the effectiveness of detection, reliability across varying patterns of motion and relevance in real time monitoring of elderly healthcare. The ultimate assessment will outline the practicality of the hip-worn embedded design to wearable fall and immobility detection.

# 4. HIGH-LEVEL SYSTEM ARCHITECTURE

## 4.1 Overview of the Proposed Component

The component proposed is a wearable belt system (i.e. hip-worn) that can identify the presence of fall incidents and post-fall immobility in older participants with inertial motion sensors and onboard processing. The design of the architecture revolves around the utilization of an MPU6050 sensor to measure body movement and an ESP32 microcontroller to perform the processing of the motion data and implement the fall detection logic in real-time. The design can be employed as a continuous wearable monitoring device as inertial sensing has been extensively applicable in fall detection studies, and embedded processing provides practical and low-latency support [2], [9], [12].

The component is created as a subset of a bigger elderly monitoring system, yet its inner structure is concentrated on motion-detected occasion recognition. The system has a systematic flow that starts with motion acquisition, which is then succeeded by feature extraction, fall verification, and generation of an alert. A hip position is significant as it enables the device to record movement patterns around the centre of mass of the body which better records the impact, position transition, and silence following a fall [8], [9].

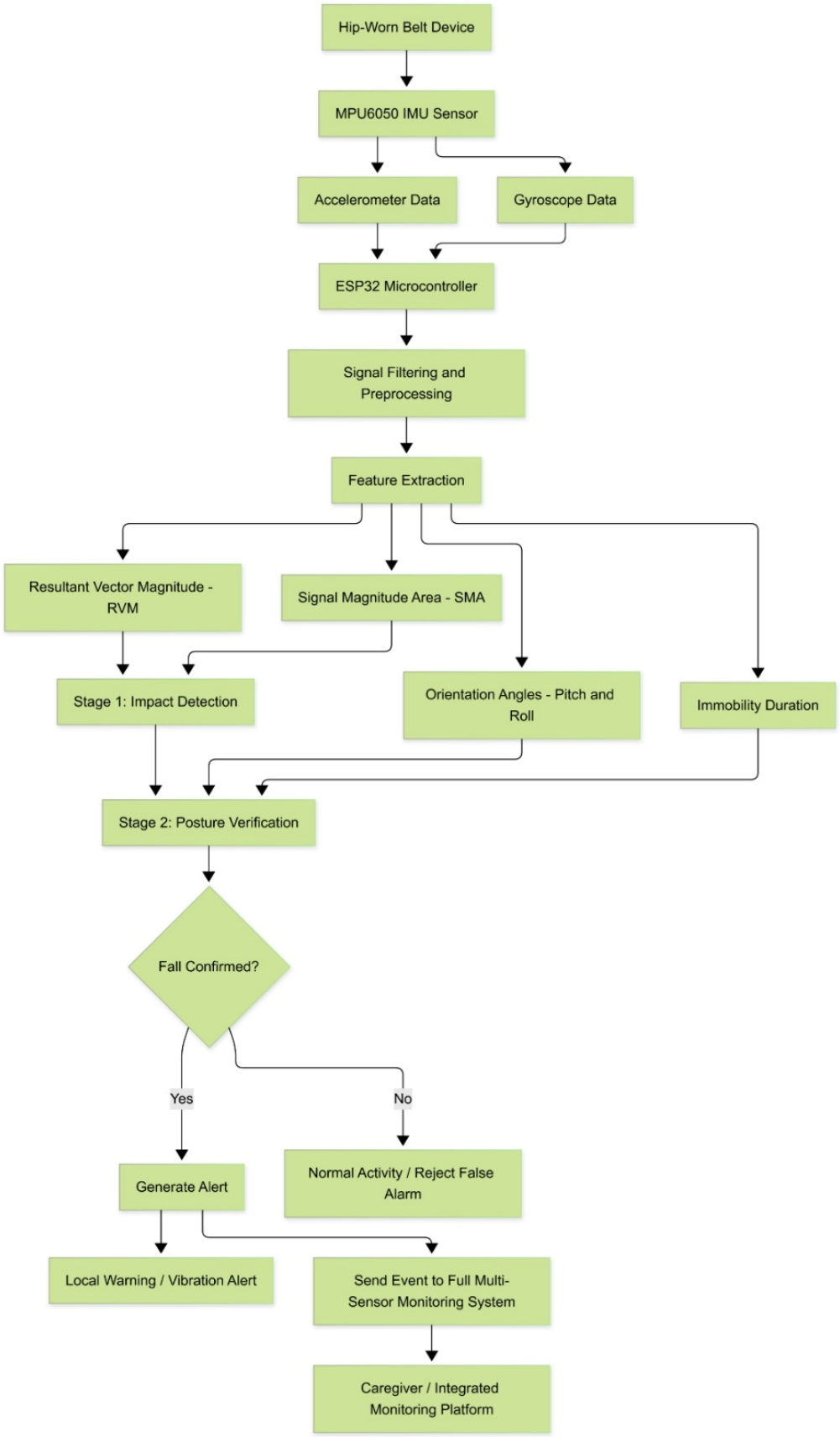



***Diagram 1- High level Architecture diagram***

*Table 1: Overview of the Proposed Component Architecture*

| Element | Description |
|---|---|
| Wearable form | Hip-worn belt device |
| Main purpose | Detect falls and post-fall immobility in elderly users |
| Input data | Tri-axial accelerometer and gyroscope signals |
| Processing unit | ESP32 microcontroller |
| Sensor module | MPU6050 IMU |
| Detection logic | Two-stage fall and immobility detection |
| Output | Local alert / integrated system alert |
| System role | Motion safety monitoring component within the full elderly monitoring system |

## 4.2 Hip-Worn Belt Architecture

The belt design to be installed on the hips is aimed at creating stable sensor positions, sound movement recordings, and convenient daily usage among elderly individuals. The MPU6050 sensor, ESP32 processing unit, the power source, and the output interface necessary to produce the alert are included in the belt. The sensor reads body motions continuously, and the microcontroller reads data, computes features, decision-making, and communicates with the rest of the monitoring system.

The positioning of the body is suitable, as hip and waist positioning has proved to be valid in wearable fall detection, as they are the nearest to the trunk motion and body balance variations [8], [9]. A belt-based structure can be used to enhance measurement consistency and contribute to long-term practical use compared to loosely attached or inconsistent placements. The architecture thus comprises both physical wearability and functional sensing.

**Table 4.2: Hip-Worn Belt Architecture Components**

| Component | Function |
|---|---|
| MPU6050 IMU | Captures acceleration and angular velocity signals |
| ESP32 | Processes sensor data and runs detection logic |
| Belt housing | Secures the device at the hip position |
| Power source | Supplies operating power to the wearable unit |
| Alert interface | Generates fall notifications or passes alerts to the integrated system |
| Communication link | Supports data sharing with the full multi-sensor platform |

## 4.3 Data Flow Between Sensors and the Detection

The sensor to detect data flow starts with the acquisition of tri-axis acceleration and gyro sensor signals at MPU6050. These crude motor motion signals are sent to the ESP32, which filters and extracts features. The stage of feature extraction computes motion-related metrics like acceleration magnitude, movement intensity, orientation change, and immobility time that are typically deployed in wearable fall detection systems [9], [10].

Once the features are extracted, the processed data is entered into the two-stage detection mechanism. The system recognizes abrupt patterns of motion that could cause a fall in the first stage. The second stage of the system cheques whether the detected event is preceded by the abnormal posture and restricted movement over a specific time. This flow aids in differentiating actual fall events and everyday activities and contributes to the more appropriate performance of detection [2], [11], [13].

***Table 3: Sensor-to-Detection Data Flow***

| Stage | Input | Process | Output |
|---|---|---|---|
| Motion sensing | Body movement | Capture accelerometer and gyroscope data | Raw motion signals |
| Signal handling | Raw motion signals | Filtering and preparation | Cleaned motion data |
| Feature extraction | Cleaned motion data | Computer RVM, SMA, orientation, immobility features | Motion feature set |

| | | | |
|---|---|---|---|
| Stage 1 detection | Motion feature set | Detect impact-related event | Suspected fall event |
| Stage 2 verification | Suspected fall event | Check posture and immobility | Confirmed fall / rejected event |
| Alert generation | Confirmed fall | Trigger local or integrated alert | Notification output |

## 4.4 Detection Output and Alert Mechanism

Upon confirmation of the fall, the system displays an alert of a suspected emergency. The alert system can be configured to assist in a quick reaction by either initiating a local alarm or sending the fall condition to the larger monitoring system. Quick and reliable alerts are imperative in a real elderly care experience since a delayed response can complicate the risk of post-fall complications [3], [7].

The motion impact does not trigger an immediate alert output. Rather, it is produced upon the second-stage verification of abnormal orientation and post fall immobility. This design consideration is significant since most everyday tasks may create sudden movement that is not a fall [2], [9]. Thus, the false-positive reduction strategy of the proposed architecture is directly connected to the alert mechanism.

## 4.5 Interaction with the Full Multi-Sensor System

Though the given proposal is dedicated to the fall and immobility detection aspect of Lakshan, the architecture is designed to be used as a component of the bigger Smart Multi-Sensor Wearable Real-Time Elderly Health Anomaly and Fall Detection system. The hip-worn belt provides kinematic safety information within the platform at large, and other modules physiological information include cardiovascular, respiratory, thermal, and neurological signals. The integration facilitates a more holistic elderly monitoring solution because it incorporates both physical-event detection and health-status awareness.

The suggested component is thus not a standalone wearable device but a specialized subsystem capable of exchanging event outputs with the entire monitoring system. Its primary input is the

detection of falls and hazardous immobility conditions and can be used alongside other sensor-based anomaly sensors. This form of integrated architecture is not isolated, but it is aligned with the overall direction of wearable, embedded, and real-time elderly healthcare systems [3], [4], [12].

***Table 5: Role of Lakshan's Component in the Full Multi-Sensor System***

| **System module** | **Main monitoring focus** | **Wear position** | **Contribution to full system** |
|---|---|---|---|
| Lakshan's component | Fall and immobility detection | Hip belt | Detects physical safety events |
| Cardiovascular module | Heart-related anomaly monitoring | Hand band | Detects cardiovascular abnormalities |
| Respiratory and thermal module | Breathing and body temperature warning | Hand band | Detects respiratory or thermal anomalies |
| Neurological stress module | Stress-related physiological state monitoring | Hand band | Detect stress or discomfort-related patterns |

In general, the high-level architecture of the proposed components will be wearable, low-cost, embedded and integration ready. The architecture will enable fall and immobility detection in real-time through hip-based motion detection, the provision of staged decisions, and triggering alerts, and can be fully integrated with the broader multi-sensing elderly health system [2], [3], [12].

# 5. USER REQUIREMENTS

In this section, the user requirements of the proposed hip-worn kinematic fall and immobility detection component are defined. The requirements must consider not just the technical performance, but the usability, comfort, response time, and compatibility with the larger multi-sensor monitoring platform since the system is to be used by the elderly. Previous studies indicate that wearable fall detection devices are best applied when they are convenient to wear in everyday life, can eliminate false alarms, and can be operational in real-life conditions (where continuous monitoring is necessary) [1], [2], [4], [13].

## 5.1 Stakeholder Identification

The three primary stakeholder groups are associated with the proposed component. The primary end users are elderly users because the system is meant to observe their physical safety and identify falls or immobilization during their daily routines. Indirect users include relatives and carers who gain access to timely alerts and safety warnings when an elderly individual might need any aid. Another stakeholder group is also healthcare providers since trustworthy fall-related information can contribute to monitoring, follow-up care, and clinical decision-making in elderly health management [1], [3], [4].

***Table 6: Stakeholder Identification***

| **Stakeholder** | **Role in the System** | **Main Need** |
|---|---|---|
| Elderly users | Primary wearer of the hip-belt device | Safe, comfortable, and reliable fall monitoring |
| Family caregivers | Receivers of fall-related alerts | Timely notification and awareness of emergency events |
| Healthcare providers | Professional monitors or advisors | Useful and dependable fall-related monitoring information |

## 5.2 Functional Requirements

The functional requirements are what the system is required to perform to accomplish its intended purpose. The suggested component should be able to detect motion of the body continuously, detect potential fall incidents, confirm post-fall immobility, and generate an alert in the case of a dangerous situation. It should also be capable of communicating with the larger team system that allows events related to falls to be integrated with other health monitoring modules. The requirements are consistent with wearable fall detector studies which highlight motion sensing, event detection, and real-time notification [2], [9], [10], [12].

***Table 7: Functional Requirements***

| ID | Functional Requirement | Description |
|---|---|---|
| FR1 | Real-time motion sensing | The system shall continuously capture tri-axial accelerometer and gyroscope data from the hip-worn MPU6050 sensor. |
| FR2 | Fall event detection | The system shall identify potential fall events based on sudden changes in motion and impact-related kinematic features. |
| FR3 | Immobility verification | The system shall verify whether the detected event is followed by abnormal posture and a period of immobility. |
| FR4 | Alert generation | The system will generate an alert when a fall with post-event immobility is confirmed. |
| FR5 | Communication with overall team system | The system shall share confirmed fall event outputs with the broader multi-sensor elderly monitoring platform. |

## 5.3 Non-Functional Requirements

The non-functional requirements explain the quality features required of the proposed component. Because it is an elderly monitoring tool that is worn around, it should be accurate to distinguish between falls, daily activities, fast enough to respond in real time and be comfortable enough to wear on a regular basis. Moreover, the system must be dependable under normal operating conditions and be economical in power consumption to facilitate feasible wearable functioning [2], [3], [11], [12].

**Table 8: Non-Functional Requirements**

| ID | Non-Functional Requirement | Description |
|---|---|---|
| NFR1 | Accuracy | The system should provide dependable fall detection performance while minimizing misclassification of normal daily activities [10], [11]. |
| NFR2 | Low latency | The system should process motion data and produce alerts with minimal delay to support timely intervention [3], [12]. |
| NFR3 | Wearability and comfort | The hip-belt device should be lightweight, stable, and comfortable for elderly users during daily activities [1], [4]. |
| NFR4 | Reliability | The system should operate consistently under normal day-to-day movement conditions and reduce unnecessary false alarms [2], [13]. |
| NFR5 | Low power operation | The device should support energy-efficient embedded operation suitable for wearable use [3], [12]. |

## 5.4 Hardware Requirements

The hardware requirements specify the physical hardware that is required to enliven the proposed system. The device needs an inertial sensor to capture motion, a local processor embedded with controller, a wearable belt housing to ensure that it is well positioned at the hip, and a source of power to drive constant activity. These hardware components enhance the proposed low-cost, real-time fall and immobility detection architecture [8], [9], [12].

***Table 9: Hardware Requirements***

| ID | Hardware Requirement | Purpose |
|---|---|---|
| HR1 | MPU6050 IMU sensor | To capture tri-axial acceleration and gyroscope data for fall-related motion analysis |
| HR2 | ESP32 microcontroller | To process sensor data and execute the two-stage fall detection logic |

| | | |
|---|---|---|
| HR3 | Hip-worn belt enclosure | To securely position the module near the body's center of mass |
| HR4 | Power supply / battery | To provide continuous power for wearable operation |
| HR5 | Alert output interface | To support local or integrated fall notification generation |

## 5.5 Software Requirements

Software requirements are a description of the programming and processing environment required to develop, test, and implement into the embedded environment. The system will need sensor reading and decision logic embedded firmware, and signal analysis, feature extraction, and evaluation data processors. The chosen software platform must be compatible with the prototype creation and real-time execution on the ESP32 platform [10], [12].

**Table 10: Software Requirements**

| ID | Software Requirement | Purpose |
|---|---|---|
| SR1 | Embedded programming environment | To develop and upload firmware to the ESP32 |
| SR2 | Sensor interfacing libraries | To read motion data from the MPU6050 reliably |
| SR3 | Signal processing scripts | To preprocess motion data and support feature extraction |
| SR4 | Algorithm implementation code | To develop the two-stage fall and immobility detection method |
| SR5 | Testing and evaluation tools | To assess detection performance using benchmark datasets and prototype outputs |

Overall, these user requirements specify the operational, technical, and usability features of the proposed component. The requirements offer a clear background to the implementation of a hip-worn wearable fall and immobility detection system used to monitor the elderly by meeting stakeholder needs, fundamental functional behaviour, and feasible system constraints [1], [3], [12].

# 6. DESCRIPTION OF PERSONNEL AND FACILITIES\

## 6.1 Personnel

### *6.1.1 Student Details*

It is a part of the research conducted by Lakshan N.K.T. (IT22196156) as a single participant in the bigger multi-sensor elderly monitoring project. The student is to create the Kinematic Fall and Immobility Detection module, which will be realized in the form of a hip-worn wearable belt. The module is aimed at the detection of fall-related body motion and post-fall inactivity utilizing inertial sensing and embedded processing. The project is aligned to the overall project goal of enhancing the safety of the elderly via real-time wearable monitoring [2], [3].

### *6.1.2 Assigned Role*

The student researcher role is assigned to design, implement, and appraise the fall and immobility detection aspect of the system. This position involves choosing the sensing method, processing inertial motion data, creating detection logic and connecting the module with the broader elderly monitoring system. As wearable fall detection requires a lot of motion interpretation, analyzing benchmark data and finding appropriate ways to differentiate falls and everyday human behaviour is also part of the role [5], [6], [11].

### *6.1.2 Responsibilities of the Student Researcher*

Technical development and documentation of the component is the work of the student researcher. Its primary task is to develop a wearable hip-belt prototype capable of detecting movement, working with accelerator and gyroscope data, and detecting falls with post-incident immobility. Moreover, the researcher needs to test the system with applicable datasets and test procedures, and make sure the component is useful to the overall multi-sensor system. These

tasks align with the previous work on fall detection focus on designing datasets, feature-based motion detection, and real-time wearing [9], [10], [12].

***Table 11: Personnel Description***

| Item | Description |
|---|---|
| Student Name | Lakshan N.K.T. |
| Student ID | IT22196156 |
| Research Component | Kinematic Fall and Immobility Detection |
| Assigned Role | Developer and researcher for the hip-worn fall detection module |
| Main Contribution | Design, implementation, testing, and documentation of the component |
| Team Role | Contributes motion-based safety monitoring to the full multi-sensor elderly healthcare system |

***Table 12: Responsibilities of the Student Researcher***

| Responsibility Area | Description |
|---|---|
| Literature review | Review academic studies on elderly falls, wearable IMU sensing, and fall detection methods [1], [2], [4] |
| Hardware development | Integrate the MPU6050 sensor and ESP32 into a hip-worn belt prototype |
| Data processing | Capture and process accelerometer and gyroscope signals for motion analysis [9], [10] |
| Algorithm design | Develop the two-stage fall and immobility detection method |
| Validation | Evaluate the component using SisFall, KFall, and related testing procedures [5], [6] |
| Documentation | Prepare proposal content, diagrams, tables, and technical descriptions for the research report |

## 6.2 Facilities and Resources

### *6.2.1 Development Environment*

Software and embedded programmed tools are part of the development environment of this component needed to design prototypes, signal handling, and testing. ESP32 firmware may then be generated in the Arduino IDE, or some other embedded development board environment, and Python may be employed to run offline motion data analysis, preprocessing, feature extraction, and performance analysis. This combination is correct since the project involves both embedded implementation and data-oriented analysis in design of fall detection [10], [12].

### *6.2.2 Hardware Resources*

The primary hardware that will be needed to complete the component is the MPU6050 inertial measurement unit, ESP32 microcontroller, hip-worn belt enclosure, battery or portable power supply, wiring accessories, and a computer to develop and test the component. These assets contribute to the purchase of motion data, integrated performance of the detection approach, and viable wearable implementation. Previous research has demonstrated that accelerator and gyroscopes based wearable sensors are highly applicable in fall detection devices within the proximity of the hip or waist [8], [9].

### *6.2.3 Data and Test Resources*

The component will consider the use of SisFall and KFall as the primary benchmark data to develop and validate the algorithm. These data sets offer labelled fall and activities of daily living inertial motion records and are therefore very useful in testing the detection performance [5], [6]. Besides, the interpretation of sensitivity, false alarms, and practical behaviour of system will be supported with associated literature and evaluation techniques of previous works [1], [10], [13]. Resources at prototype level will also be a controlled development environment to verify that ESP32-based wearable module works as required under experimental motion conditions.

*Table 13: Facilities and Resources*

| Category | Resource | Purpose |
|---|---|---|
| Development environment | Arduino IDE / embedded programming tools | To develop and upload firmware to ESP32 |
| Development environment | Python environment | To perform preprocessing, feature extraction, and testing |
| Hardware resource | MPU6050 IMU sensor | To capture accelerometer and gyroscope data |
| Hardware resource | ESP32 microcontroller | To execute embedded fall detection logic |
| Hardware resource | Hip-belt enclosure | To secure the wearable module at the hip position |
| Hardware resource | Battery / power source | To support portable system operation |
| Hardware resource | Personal computer / laptop | To run development, testing, and documentation tasks |
| Dataset resource | SisFall dataset | To support fall detection analysis and benchmarking [5] |
| Dataset resource | KFall dataset | To support validation using labeled inertial motion data [6] |
| Testing resource | Controlled prototype testing setup | To observe basic system behavior and response during trial movement scenarios |

*Table 14: Expected Test Cases*

| Test Case ID | Test Scenario | Input Condition | Expected System Response | Expected Result |
|---|---|---|---|---|
| TC01 | Normal walking | User walks normally while wearing the hip belt | System captures motion data continuously without generating an alert | No fall detected |

| | | | | |
|---|---|---|---|---|
| TC02 | Sitting down normally | User sits on a chair with normal body movement | System processes motion change but do not classify it as a fall | No alert generated |
| TC03 | Lying down intentionally | User lies down slowly on a bed or flat surface | System detects posture change but rejects fall classification | No alert generated |
| TC04 | Sudden bending movement | User bends quickly to pick up an object | System records rapid motion but does not confirm a fall | False alarm avoided |
| TC05 | Simulated fall with impact | User performs a controlled fall scenario with sudden impact | System identifies abnormal impact pattern | Suspected fall detected |
| TC06 | Fall followed by immobility | Simulated fall occurs and user remains motionless for a defined period | System confirms fall and immobility | Alert generated |
| TC07 | Fall followed by quick recovery | Simulated fall occurs but user stands up immediately | System detects impact but rejects final fall confirmation after recovery | No final alert or temporary verification only |
| TC08 | Rapid sitting mistaken as fall | User sits abruptly with high acceleration | System enters verification stage and checks posture/immobility | False alert prevented |
| TC09 | Device worn correctly on hip | Belt is positioned properly at the hip during movement | Sensor captures stable motion signals for analysis | Accurate data acquisition |
| TC10 | Device worn incorrectly | Belt is loose or placed off position | Motion signals become unstable or inconsistent | Reduced performance / adjustment required |

| | | | | |
|---|---|---|---|---|
| TC11 | Continuous daily activity monitoring | User performs mixed normal activities over time | System continues sensing and processing in real time | Stable operation without unnecessary alerts |
| TC12 | Confirmed fall event sent to team system | Fall and immobility are confirmed | System generates local alerts and shares events with integrated monitoring system | Emergency notification delivered |

In general, the staff, facilities, and resources listed in this section are the basis needed to build and assess the component of Lakshan. A well-defined student role, the appropriate embedded hardware and the apt software tools, together with a set of benchmark data sets are conducive to the creation of a realistic, hip-worn fall and immobility detection module in elderly healthcare monitoring [2], [5], [6], [12].

# 7. COMMERCIALIZATION PLAN

The suggested hip-worn fall and immobility detection device is highly marketable due to falls being one of the most significant safety menaces facing senior citizens, particularly individuals who live alone or need constant care [1], [3]. This need can be fulfilled through a wearable belt-based solution that will provide a low-cost, practical, and real-time monitoring device that will assist a faster response in case of an emergency involving a fall. A hip-worn design is commercially viable compared to more complex monitoring systems since it is centered around a clear and pressing use-case and offers ease of wear and a consistent motion capture [8], [9].

The main target market will consist of the elderly population, family caregivers, assisted living facilities, elderly care facilities, and home healthcare service providers. These user groups need cost effective safety technologies which can be deployed in day-to-day situations without expensive infrastructural support. The suggested element can be framed as an intelligent old age safety device that enhances confidence, promotes self-sufficiency, and minimizes the danger of falling without help [2], [4]. It has a value proposition of integrating real-time motion

sensing, fall event detection and post-fall immobility verification within one wearable solution. The significance of this is in the fact that systems identifying sudden motion only can generate false alarms, whereas a combined verification method can enhance the practical reliability [11], [13].

On the business side, the device may be designed either as a low-priced single device or as a component of a more comprehensive multi-sensor elderly monitoring system. It would be possible to sell devices directly to individuals and institutions and generate revenue or offer bundled healthcare monitoring options in collaboration with elder care providers. As the system relies on widely accessible hardware like the ESP32 and MPU6050, the cost of production can be kept relatively affordable and supports both competitive pricing and affordability. The commercialization potential is enhanced in the long run by the rising demand of wearable elderly care technologies, and the rising demand of preventive and home-based monitoring solutions [3], [12].

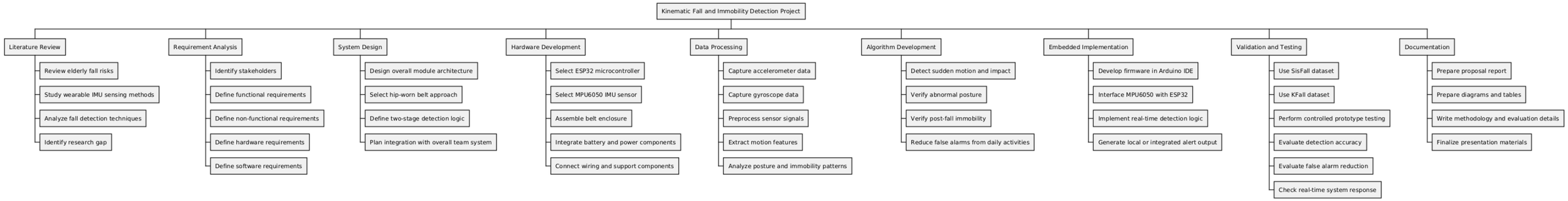
# 8. BUDGET AND JUSTIFICATION\

***Table 14: Budget and Justification***

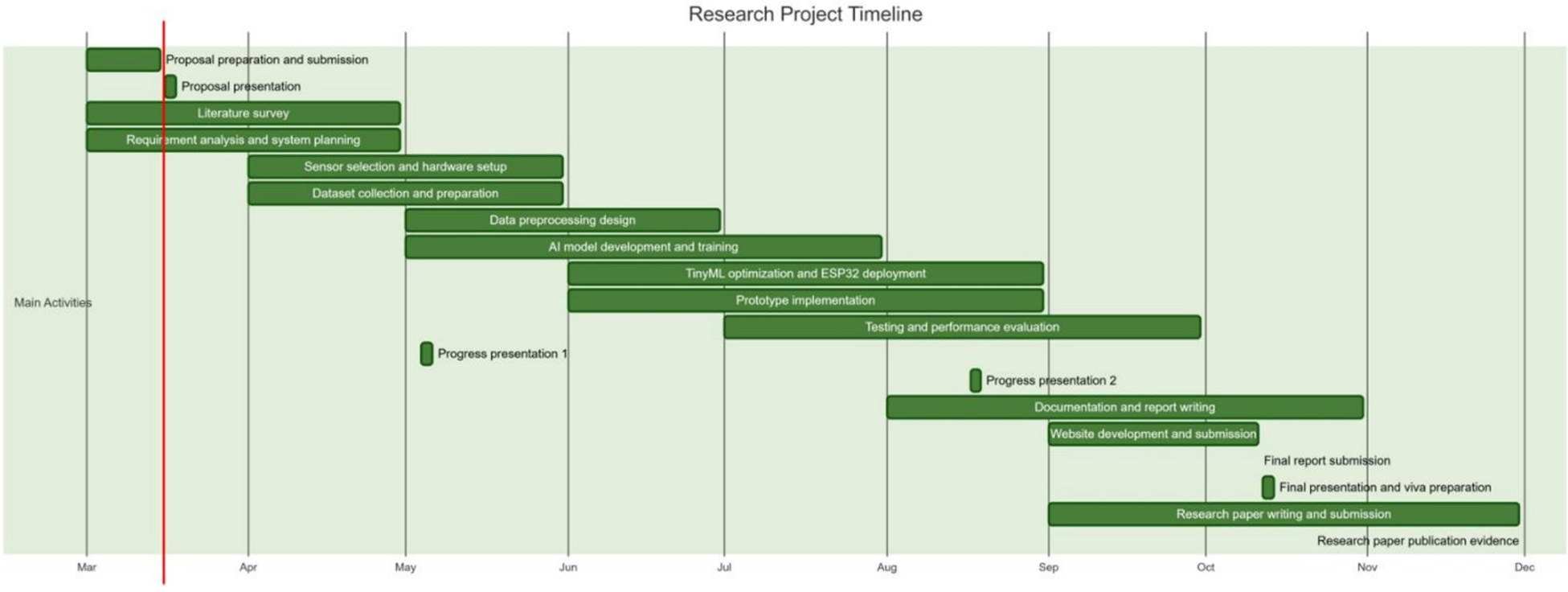
| Cost Category | Item | Estimated Cost (LKR) | Justification |
|---|---|---|---|
| Hardware | ESP32 microcontroller | 3,500 | Required as the main embedded processing unit to read sensor data and execute the fall detection logic in real time. |
| Hardware | MPU6050 IMU sensor | 1,500 | Required to capture tri-axial accelerometer and gyroscope data for motion sensing and fall analysis. |
| Hardware | Belt enclosure and wearable accessories | 2,000 | Needed to secure the prototype at the hip position and support practical wearable testing. |

| | | | |
|---|---|---|---|
| Hardware | Battery / power components | 2,000 | Required to power the wearable module during standalone operation and prototype testing. |
| Hardware | Wires, connectors, breadboard / assembly items | 1,500 | Needed for circuit integration, hardware setup, and prototype assembly. |
| Software | Arduino IDE / Python tools | 0 | Open-source tools will be used for embedded programming, data processing, and testing, so no licensing cost is required. |
| Data and Testing | Benchmark datasets (SisFall and KFall) | 0 | Publicly available research datasets will be used for development and validation, so no purchase cost is involved [5], [6]. |
| Development Effort | Prototype development and testing support | 8,000 | Represents the estimated value of time and effort spent on implementation, debugging, algorithm testing, and system refinement. |
| Documentation | Printing, report preparation, and presentation materials | 2,500 | Required for proposal/report submission, diagrams, and presentation-related preparation. |
| Contingency | Miscellaneous technical adjustments | 2,000 | Reserved for unexpected component replacement, small modifications, or additional testing needs. |
| | **Total Estimated Cost** | **23,000** | |

## 9. WORK BREAKDOWN STRUCTURE (WBS)




## 10. GANTT CHART



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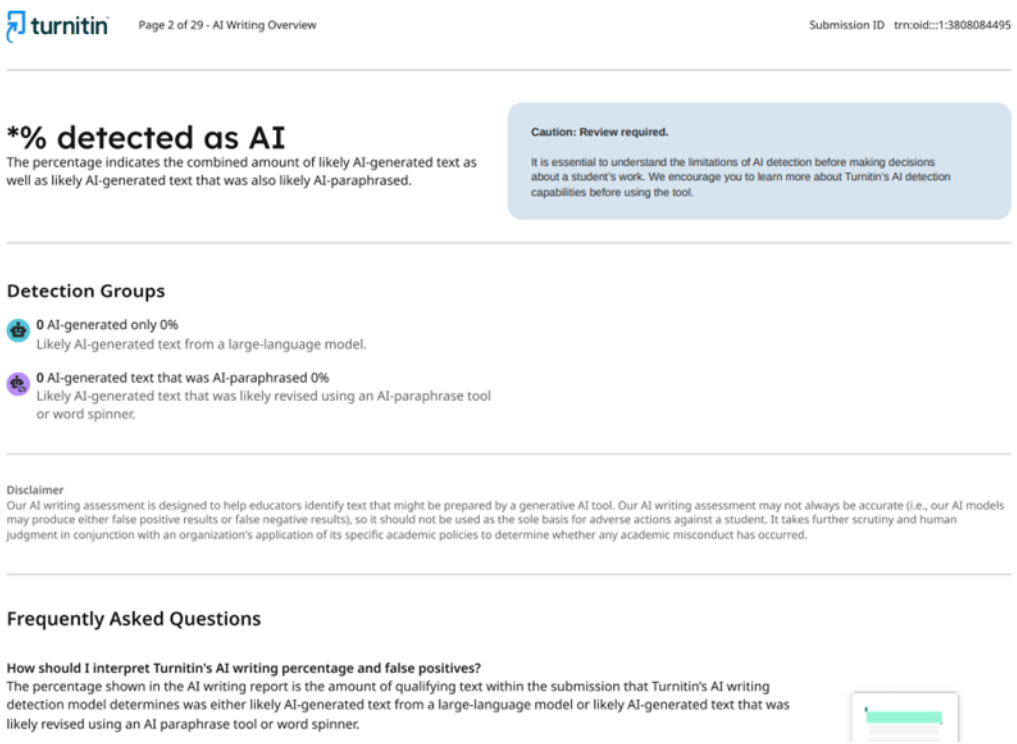
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